3.1.7 Oxidation, reduction and redox equations: Sheet 2

Writing half equations · 15 marks · about 25 minutes

Key information

- A half equation shows either the oxidation or the reduction, with electrons on one side.
- To balance in acidic solution: (1) balance the main atom; (2) add H_2O to balance O; (3) add H^+ to balance H; (4) add e^- to balance charge.
- Electrons appear on the **left** for a reduction and on the **right** for an oxidation.
- Check: atoms and total charge must be the same on both sides.

Section A: Self-quiz

01 State, in terms of electrons, the meaning of the term oxidising agent.

[1 mark]

02 Which incomplete half-equation is balanced by adding two H^+ ions and one electron to the left-hand side?

[1 mark]

A $CH_3CHO \rightarrow CH_3CH_2OH$

B $VO^{2+} \rightarrow V^{3+} + H_2O$

C $HNO_2 \rightarrow NO + H_2O$

D $O_2 \rightarrow H_2O_2$

03 Which statement about SO_2 acting as a reducing agent is correct?

[1 mark]

A SO_2 gains electrons and the oxidation state of sulfur decreases.

B SO_2 gains electrons and the oxidation state of sulfur increases.

C SO_2 loses electrons and the oxidation state of sulfur decreases.

D SO_2 loses electrons and the oxidation state of sulfur increases.

04 Which equation does not represent a redox reaction?

[1 mark]

A $Mg + 2\,HCl \rightarrow MgCl_2 + H_2$

B $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$

C $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

D $CuO + 2\,HCl \rightarrow CuCl_2 + H_2O$

05 Which of these is not a redox reaction?

[1 mark]

A $Cu_2O + H_2SO_4 \rightarrow CuSO_4 + Cu + H_2O$

B $MgO + 2HCl \rightarrow MgCl_2 + H_2O$

C $SnCl_2 + HgCl_2 \rightarrow Hg + SnCl_4$

D $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

Section B: Exam questions

06 When an acidified solution of sodium nitrite ($NaNO_2$) is added to aqueous potassium iodide, iodine and nitrogen monoxide (NO) are formed.

06.1 Write a half-equation for the conversion of NO_2^- in an acidic solution into NO

[1 mark]

06.2 Write a half-equation for the conversion of I^- into I_2

[1 mark]

07 When chlorine is added to water, there is a redox reaction, as shown by the equation

$Cl_2 + H_2O \rightleftharpoons HClO + HCl$

Give two half-equations to show the oxidation and reduction processes that occur in this redox reaction.

[2 marks]

08 Iodide ions can be oxidised to iodine using $Cr_2O_7^{2-}$ ions. State symbols are not required.

08.1 Deduce a half-equation to show the oxidation of iodide ions to iodine.

[1 mark]

08.2 Deduce a half-equation for the conversion in acidic solution of $Cr_2O_7^{2-}$ ions to Cr^{3+} ions.

[1 mark]

09 Deduce the half-equation, in acidic conditions, for the conversion of ClO_3^- into Cl^-

[1 mark]

10 In a methanol–oxygen fuel cell, the overall reaction is

$CH_3OH(l) + 1½O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

10.1 At the positive electrode, oxygen reacts with hydrogen ions to form water. Give a half-equation for this reaction.

[1 mark]

10.2 At the negative electrode, methanol reacts with water to produce carbon dioxide and hydrogen ions. Give a half-equation for this reaction.

[1 mark]

11 In a glucose–oxygen fuel cell, glucose ($C_6H_{12}O_6$) molecules react with water at the negative electrode to form carbon dioxide and hydrogen ions.

Deduce the half-equation for the reaction at the negative electrode.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	An electron acceptor	Source: 7404/1 Jun 2024 Q09.1 Do not accept electron pair acceptor or gain of electrons	1
02	**B**	Source: 7404/1 Jun 2023 Q18	1
03	**D**	Source: 7404/1 Jun 2025 Q20	1
04	**D**	Source: 7404/1 Jun 2019 Q15	1
05	**B**	Source: 7404/1 Jun 2017 Q20	1
06.1	$NO_2^- + e^- + 2H^+ \rightarrow NO + H_2O$	Source: 7404/1 Jun 2017 Q08.2–08.3 Or double	1
06.2	$2I^- \rightarrow I_2 + 2e^-$	Or half	1
07	Oxidation: $Cl_2 + 2H_2O \rightarrow 2HClO + 2H^+ + 2e^-$ Reduction: $Cl_2 + 2H^+ + 2e^- \rightarrow 2HCl$	Source: 7404/1 Jun 2018 Q07.2 Allow $Cl_2 + 2H_2O \rightarrow 2ClO^- + 4H^+ + 2e^-$; allow $Cl_2 + 2e^- \rightarrow 2Cl^-$ Accept – $2e^-$ on the other side If both equations correct but incorrect order, allow 1 Ignore state symbols	2
08.1	$2I^- \rightarrow I_2 + 2e^-$	Source: 7404/1 Jun 2019 Q03.5–03.6 Allow multiples / ignore ss	1
08.2	$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	Allow multiples / ignore ss	1
09	$ClO_3^- + 6H^+ + 6e^- \rightarrow Cl^- + 3H_2O$	Source: 7404/1 Jun 2025 Q02.6 Allow multiples	1
10.1	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	Source: 7405/1 Jun 2023 Q08.1–08.2	1
10.2	$CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$		1
11	$C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 24H^+ + 24e^-$	Source: 7405/1 Nov 2020 Q11.1 Accept multiples	1